

**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 10 August 2026, 10:27 AM**State** Finished**Completed on** Monday, 10 August 2026, 10:40 AM**Time taken** 13 mins 30 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**For example:**

| Input | Result |
|-------|--------|
| 9     | 12     |

**Answer:** (penalty regime: 0 %)

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```
#include <stdio.h>

int main()
{
    int n;
    int count = 0;

    scanf("%d", &n);

    int i = 1;
    count++;

    int s = 1;
    count++;

    while (s <= n)
    {
        count++;
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 9     | 12       | 12  | ✓ |
| ✓ | 4     | 9        | 9   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 10 August 2026, 10:32 AM**State** Finished**Completed on** Monday, 10 August 2026, 10:40 AM**Time taken** 7 mins 48 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("");
                printf("");
                break;
            }
        }
    }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      int count = 1;
7
8      scanf("%d", &n);
9
10     if (n == 1)
11     {
12         count++;
13     }
14     else
15     {
16
17         for (int i = 1; i <= n; i++)
18         {
19             count ++;
20
21             for (int j = 1; j <= n; j++)
22             {
23                 count++;
24                 count++;
25                 count++;
26
27                 break;
28
```

```
29         }  
30         count++;  
31     }  
32     count++;  
33 }  
34  
35 printf("%d", count);  
36  
37 return 0;  
38 }  
39  
40  
41
```

|   | Input | Expected | Got  |   |
|---|-------|----------|------|---|
| ✓ | 2     | 12       | 12   | ✓ |
| ✓ | 1000  | 5002     | 5002 | ✓ |
| ✓ | 143   | 717      | 717  | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 10 August 2026, 10:40 AM**State** Finished**Completed on** Monday, 10 August 2026, 10:45 AM**Time taken** 4 mins 7 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i;
6      int count = 0;
7
8      scanf("%d", &n);
9
10     for (i = 1; i <= n; ++i)
11     {
12         count++;
13         count++;
14
15         if (n % i == 0)
16         {
17             count++;
18         }
19     }
20
21     count++;
22
23     printf("%d", count);
24
25     return 0;
26 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 12    | 31       | 31  | ✓ |
| ✓ | 25    | 54       | 54  | ✓ |

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 4     | 12       | 12  | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 10 August 2026, 10:45 AM**State** Finished**Completed on** Monday, 10 August 2026, 10:48 AM**Time taken** 3 mins 12 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, j, k;
6      int c = 2;
7
8      scanf("%d", &n);
9
10     for (i = n / 2; i < n; i++)
11     {
12         c += 2;
13
14         for (j = 1; j < n; j = 2 * j)
15         {
16             c += 2;
17
18             for (k = 1; k < n; k = k * 2)
19             {
20                 c += 2;
21             }
22         }
23     }
24
25     printf("%d", c);
26
27     return 0;
28 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 4     | 30       | 30  | ✓ |

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 10    | 212      | 212 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 10 August 2026, 10:49 AM**State** Finished**Completed on** Monday, 10 August 2026, 10:50 AM**Time taken** 1 min 16 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      int c = 0;
7
8      scanf("%d", &n);
9
10     int rev = 0, remainder;
11     c++;
12
13     while (n != 0)
14     {
15         c++;
16
17         remainder = n % 10;
18         c++;
19
20         rev = rev * 10 + remainder;
21         c++;
22
23         n /= 10;
24         c++;
25     }
26
27     c++;
28     c++;
29
30     printf("%d", c);
31
32     return 0;
33 }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 12    | 11       | 11  | ✓ |
| ✓ | 1234  | 19       | 19  | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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